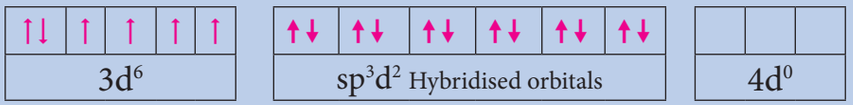
Complex	$[CoF_6]^{3-}$
Hybridised orbitals of the metal atom in the complex	
Geometry	Octahedral In this complex outer d orbitals are involved in the hybridisaion and hence the complex is called outer orbital complex
Magnetic property	No. of unparied electrons = 4; Hence paramagnetic
Magnetic moment (Using spin only formula)	$\mu_s = \sqrt{n(n+2)} = \sqrt{4(4+2)} = 4.899$ BM

Limitations of VBT

Eventhough VBT explains many of the observed properties of complexes, it still has following limitations

1. It does not explain the colour of the complex
2. It considers only the spin only magnetic moments and does not consider the other components of magnetic moments.
3. It does not provide a quantitative explanation as to why certain complexes are inner orbital complexes and the others are outer orbital complexes for the same metal. For example, $[Fe(CN)_6]^{4-}$ is diamagnetic (low spin) whereas $[FeF_6]^{4-}$ is paramagnetic (high spin).

Evaluate yourself 1:

7. The spin only magnetic moment of Tetrachloridomanganate(II)ion is 5.9 BM. On the basis of VBT, predict the type of hybridisation and geometry of the compound.
8. Predict the number of unpaired electrons in $[CoCl_4]^{2-}$ ion on the basis of VBT.
9. A metal complex having composition $Co(en)_2Cl_2Br$ has been isolated in two forms A and B. (B) reacted with silver nitrate to give a white precipitate readily soluble in ammonium hydroxide. Whereas A gives a pale yellow precipitate. Write the formula of A and B. state the hybridization of Co in each and calculate their spin only magnetic moment.

XII U5 Coordination jagan.indd 152 3/29/2019 6:28:03 PM